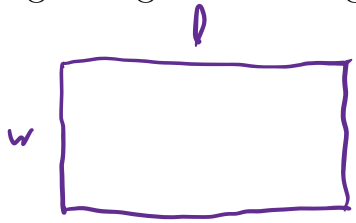
**Problem 7.** TIER 1 — TWO RATES AT ONCE

A rectangle's length is increasing at 3 cm/s while its width is decreasing at 2 cm/s. At the moment the length is 20 cm and the width is 15 cm, is the area growing or shrinking, and how fast?

1. Draw + Label
2. What & when
3. Equation
4. Differentiate w.r.t. $t$
5. Sub + Solve

(1)




(2) $\frac{dl}{dt} = 3$, $\frac{dw}{dt} = -2$

$\frac{dA}{dt} = ?$, $l = 20$, $w = 15$

(3) $A = lw$

(4) $\frac{d}{dt}A = \frac{d}{dt}(lw)$

$\frac{dA}{dt} = l\frac{dw}{dt} + w\frac{dl}{dt}$

(5) $\frac{dA}{dt} = 20(-2) + 15(3) = -40 + 45 = 5 \frac{ft^2}{sec}$